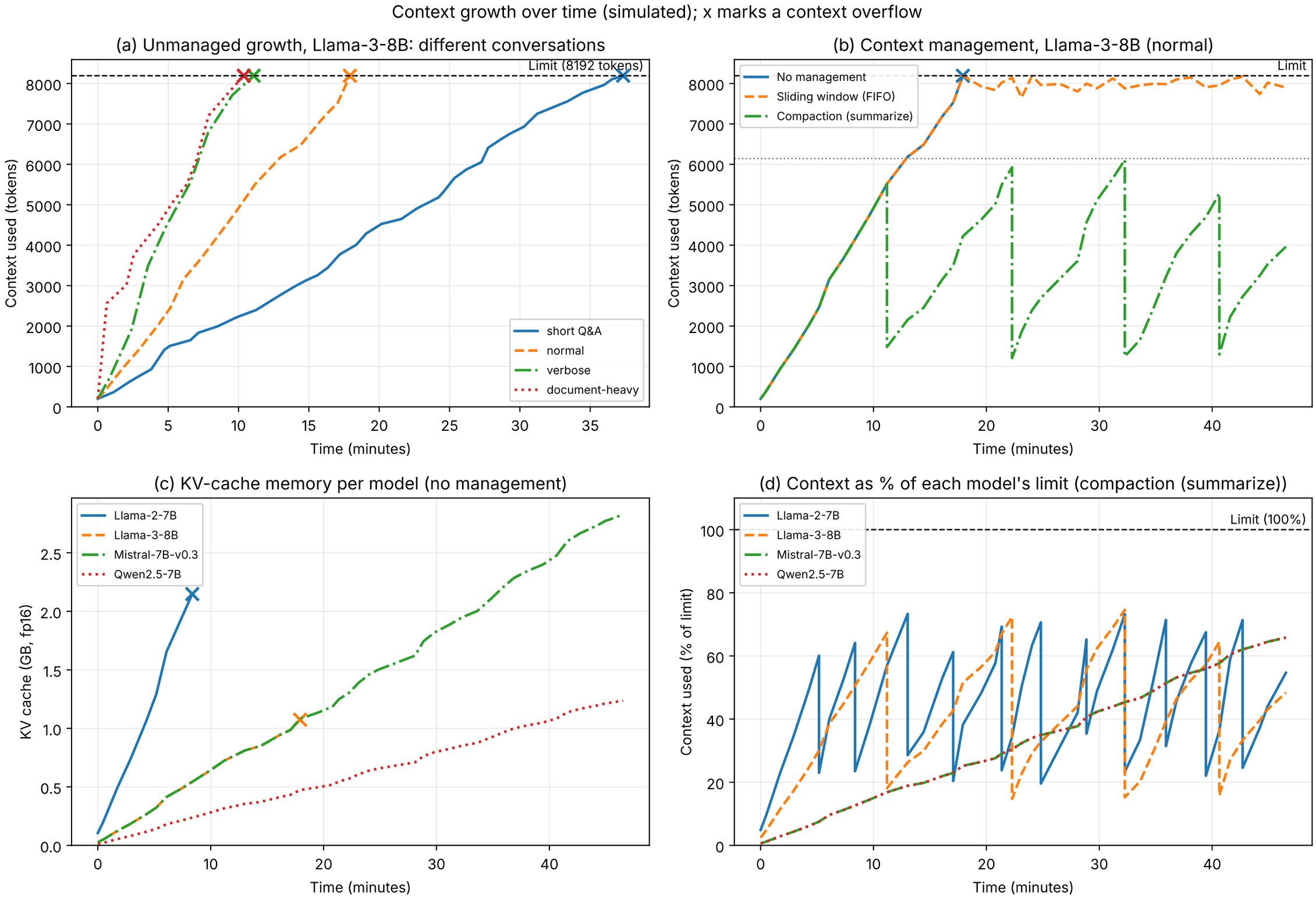
Context growth over time (simulated); x marks a context overflow
(a) Unmanaged growth, Llama-3-8B: different conversations
Limit (8192 tokens)
Context used (tokens)
Time (minutes)
short Q&A
normal
verbose
document-heavy
(b) Context management, Llama-3-8B (normal)
Limit
No management
Sliding window (FIFO)
Compaction (summarize)
Context used (tokens)
Time (minutes)
(c) KV-cache memory per model (no management)
Llama-2-7B
Llama-3-8B
Mistral-7B-v0.3
Qwen2.5-7B
KV cache (GB, fp16)
Time (minutes)
(d) Context as % of each model's limit (compaction (summarize))
Limit (100%)
Context used (% of limit)
Time (minutes)